

TDDE62-IDS-LAB2

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March 2026

1 Pings

Suggested Snort Rule:

```
alert icmp any any -> any any (msg:"ICMP Echo Request  
detected"; itype:8; sid:1000001; rev:1;)
```

This command alerts all pings from all source IPs to all destination IPs. The flag itype:8 alerts only if the ICMP type is Echo Request and ignores other ICMP messages.

To block pings completely, we suggest the following IPTABLES rule:

```
iptables -A FORWARD -p icmp -icmp-type echo-request -j DROP
```

This rule just drops all forwarded Echo Requests. To only block ping requests to our network, we could add **-d 10.19.1.0/24** before **-j DROP**.

2 FIN Scanning

Suggested Snort Rule:

```
alert tcp any any -> 10.19.1.0/24 any (msg:"Possible FIN scan  
detected"; flags:F,12; threshold:type threshold, track by _src, count  
10, seconds 10; sid:10002; rev:1;)
```

This rule alerts if there are too many incoming tcp packets that have only the FIN flag set.

To avoid such FIN Scanning packets, we can use the following IPTABLES rule:

```
iptables -A FORWARD -p tcp -tcp-flags ALL FIN -m recent -name  
finscan -seconds 10 -hitcount 10 -j DROP
```

As packets could have only the FIN flag set, we need to use a threshold. Similar to the snort rule we decided to use 10 packets within 10 seconds.

3 SYN Flood

If there was a SYN Flood from a single source IP, we could just count SYN packets from this source and limit its requests. But as the source is randomized, we have a hard time distinguishing the flood packets from legitimate SYN requests. Snort can also count packets by destination. Thus, the task is to find a suitable threshold that seems as being too high for regular mail server usage. We decided for an amount of 100 SYN packets within 10 seconds. However, this threshold and time might be adjusted in order to match the expected amount of traffic for a given mail server. Our suggested snort rule is:

```
alert tcp any any -> 10.19.1.141 110 (msg:"Possible SYN flood on  
mail server"; flags:S,12; threshold:type threshold, track by _dst,  
count 100, seconds 10; sid:10003; rev:1;)
```

One can notice, that there are suspicious header information in the flood packets: the acknowledgment number field is nonzero while the ACK flag is not set. We could use this to block such TCP packets. On the other hand, one could simply limit the amount of packets with SYN flags that are forwarded to the server, i.e. limit the amount of new connections.

4 File Inclusion Attack

Suggested snort rule:

```
alert tcp any any -> 10.19.1.140 80 (msg:"File inclusion attack -  
/etc/passwd"; content:"/etc/passwd"; sid:10004; rev:1;)
```

To alert if the passwd file at /etc is downloaded, we simply look for the file path (/etc/passwd) in the http GET request. It is not smart to alert every GET request, because it will trigger false alerts. Using the suggested rule, only access the exact file is covered. To protect more files, more rules with a more general file path like "/etc" could be used.